**GROUP WORK PROJECT # _3__** MScFE 620: Derivative Pricing
**GROUP NUMBER:** __8300_____

| FULL LEGAL NAME | LOCATION (COUNTRY) | EMAIL ADDRESS | MARK X FOR ANY NON-CONTRIBUTING MEMBER |
|---|---|---|---|
| Maximilian Berghofer | Germany | max.bghf@gmx.de | |
| MD RASHEDUL ISLAK RASEL | Bangladesh | rashedul.rasel@hotmail.com | |
| Tai Wu | United Kingdom | taiwu2022@gmail.com | |

| **Statement of integrity:** By typing the names of all group members in the text boxes below, you confirm that the assignment submitted is original work produced by the group (excluding any non-contributing members identified with an "X" above). | |
|---|---|
| **Team member 1** | **Maximilian Berghofer** |
| **Team member 2** | **MD RASHEDUL ISLAM RASEL** |
| **Team member 3** | **Tai Wu** |

| Use the box below to explain any attempts to reach out to a non-contributing member. Type (N/A) if all members contributed.<br>**Note:** You may be required to provide proof of your outreach to non-contributing members upon request. |
|---|
| |

**Step 1:**
**Team member A:**

Answer 5: Heston Model Pricing with Correlation -0.30

Using the Heston Model with a correlation of -0.30, we obtain the following option prices:

Call Price: $2.89

Put Price: $2.84

With a moderate negative correlation, the stock price and volatility are inversely related but not drastically so. This results in relatively stable option prices, as volatility does not spike excessively when the stock price moves. The option premiums reflect this balanced stochastic behavior.

Answer 6: Heston Model Pricing with Correlation -0.70

When we adjust the correlation to -0.70, meaning the relationship between stock price and volatility becomes more strongly negative, the option prices change as follows:

Call Price: $2.19

Put Price: $3.35

With a stronger negative correlation, volatility rises more significantly when the stock price drops. This leads to a lower call price because higher volatility is associated with declining stock prices, reducing the expected value of the call. Conversely, the put option becomes more expensive, as the increased likelihood of large downward price swings makes it more valuable.

Answer 7: Delta & Gamma for Both Cases

For correlation -0.30:

Delta (Call): 0.5383 → A $1 increase in stock price results in a $0.54 increase in the call price.

Delta (Put): -0.4486 → A $1 increase in stock price causes the put price to drop by about $0.45.

Gamma (Both): 0.0546 → Delta's sensitivity to stock price changes remains moderate.

For correlation -0.70:

Delta (Call): 0.4928 → The call option's delta decreases, meaning it becomes less sensitive to stock price movements due to the stronger impact of volatility.

Delta (Put): -0.4791 → The put option's delta increases, showing it reacts more strongly to stock price changes, given the higher probability of large downward moves.

Gamma (Both): 0.0577 → A slight increase in gamma, indicating delta changes more rapidly in this scenario.

**Step 1:**

**Team member B:** Jump Modeler

Answer 8.

We will use Monte Carlo simulation to get the stock price paths under the Merton Model. Given the parameter space, the ATM European call and put option with jump intensity parameter equal to 0.75. We have

Jump Intensity ($\lambda$ = 0.75 ), Call price = 2.83, Put price = 3.74

Since we use Monte Carlo simulation, here we generate 10,000 sample paths for the stock price evolution, at the end of each simulation, the call and put option are obtained by the risk0neutral discounted payoff.

Answer 9.

We repeat the process as previous, for a ATM European option by Merton Model given by

Jump Intensity ($\lambda$ = 0.25 ), Call price = 2.88, Put price = 3.02

The result indicate that a higher jump intensity $\lambda$ increase put option price and call option. Since larger jumps increase risk for both sides, which will lead to more valuable options.

While comparing both sizes of change, we find that call option prices are relatively stable, while put options increase a lot, this is because larger downward jumps increase downside risk, making puts more valuable than call.

Answer 10.

Using the finite difference approximation, we find the Delta and Gamma for both cases. Where Delta measures the sensitivity to stock price and Gamma measures the sensitivity of Delta. In both cases we find that delta is higher as jump intensity increases, since options are more reactive to stock price change when jumps occur frequently.

Jump Intensity ($\lambda$ = 0.75 ), Delta = 2.32, Gamma = -395.24

Jump Intensity ($\lambda$ = 0.25 ), Delta = -0.13, Gamma = -125.49

Meanwhile, gamma is also higher (in absolute term) for higher jump, it also follows our intuition, since higher jump likelihood, suggesting that the rate at which delta changes is more volatile.

**Team Member C:**

**Answer 11.**

**Put-Call Parity Check for Heston and Merton Models**

Put-Call Parity Formula (European Options):

C−P=S0−Ke^−rT

**Parameters:**

S0=80, K=80 (ATM), r=5.5%, T=0.25.

S0−Ke^−rT=80−80e^−0.055×0.25=80−78.92=1.08.

**1. Heston Model Results**

| Correlation | Call Price (C) | Put Price (P) | C-P | Parity Holds? |
|---|---|---|---|---|
| -0.3 | $2.89 | $2.84 | 0.05 | No (0.05 ≠ 1.08) |
| -0.7 | $2.19 | $3.35 | -1.16 | No (-1.16 ≠ 1.08) |

**Heston Model:** Stochastic volatility disrupts put-call parity. The correlation parameter introduces skewness in volatility, leading to mispricing relative to the risk-neutral framework.

**2. Merton Model Results**

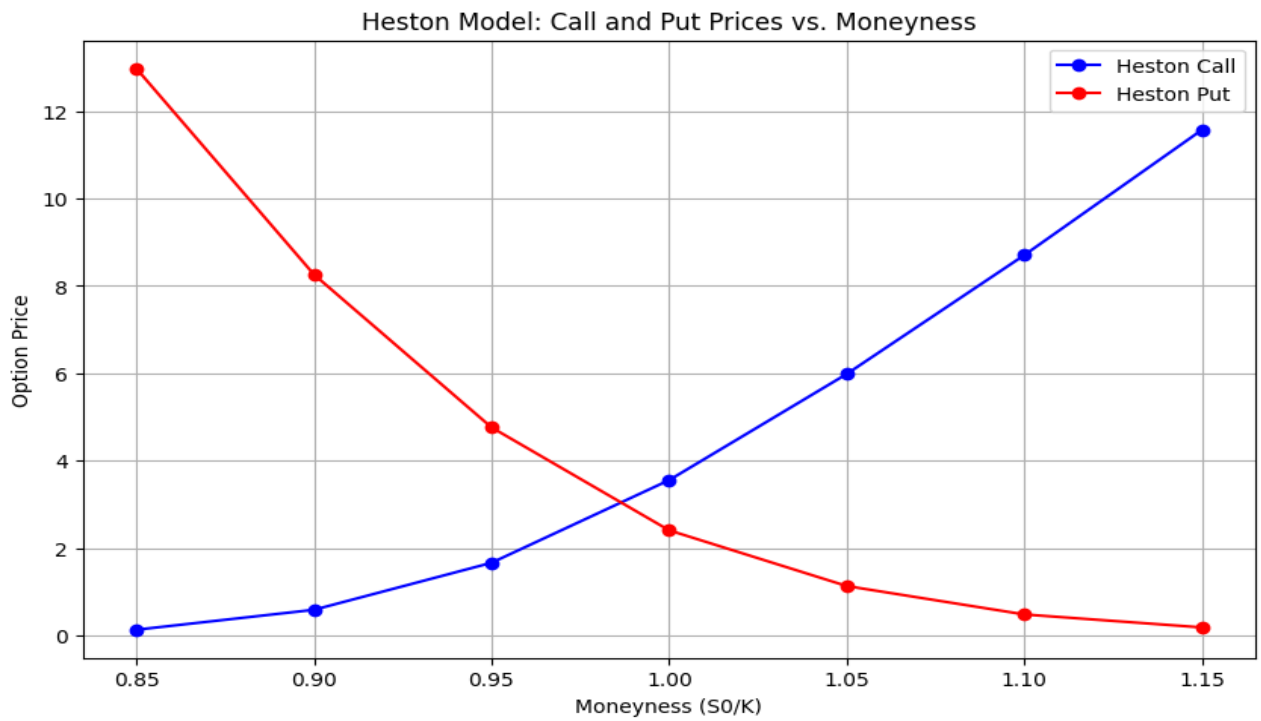
| Jump Intensity | Call Price (C) | Put Price (P) | C-P | Parity Holds? |
|---|---|---|---|---|
| 0.75 | $2.83 | $3.74 | -0.91 | No (-0.91 ≠ 1.08) |
| 0.25 | $2.88 | $3.02 | -0.14 | No (-0.14 ≠ 1.08) |

**Merton Model:** Jumps introduce non-linear payoffs and tail risk, violating put-call parity. Higher jump intensity amplifies the discrepancy.

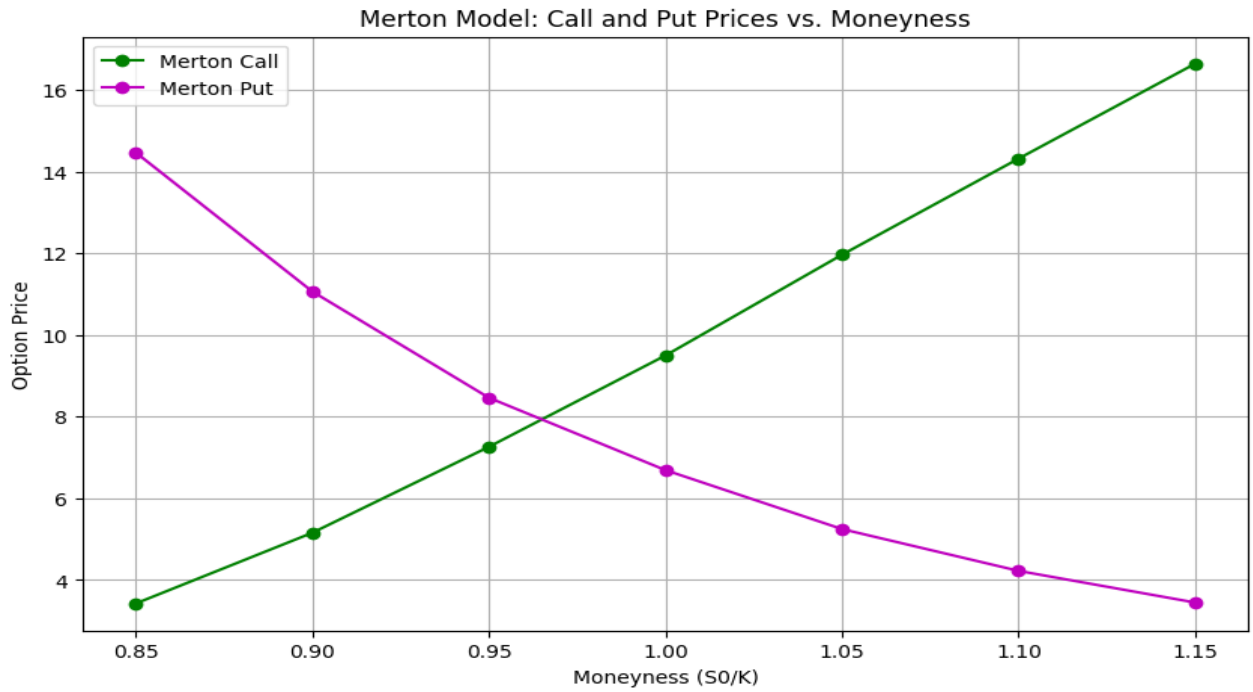
**Answer 12**

Find the Option Prices for Heston and Merton Models in the Python file:

| Strike | Moneyness | Heston Call | Heston Put | Merton Call | Merton Put |
|---|---|---|---|---|---|
| 94.12 | 0.85 | 0.14 | 12.97 | 3.44 | 14.53 |
| 88.89 | 0.9 | 0.59 | 8.27 | 5.24 | 10.98 |
| 84.21 | 0.95 | 1.67 | 4.76 | 7.32 | 8.41 |
| 80 | 1 | 3.54 | 2.43 | 9.57 | 6.64 |
| 76.19 | 1.05 | 5.98 | 1.13 | 11.87 | 5.28 |
| 72.73 | 1.1 | 8.73 | 0.48 | 14.21 | 4.23 |
| 69.57 | 1.15 | 11.55 | 0.19 | 16.65 | 3.45 |




**Trend: OTM puts and ITM calls are more expensive due to volatility skew.**




**Trend: OTM puts spike due to jump risk, while ITM calls rise from upward jumps.**

Step 2

Answer 13:

When we use the heston model with monte carlo simulation to price an american call option, we get different results compared to the european options from questions 5 and 8. the ability to exercise the option early impacts the pricing, and we observe how the correlation value affects this difference.

For correlation -0.30, the american call option is priced at 2.70$, while for correlation -0.70, the price is 2.02$.

Observations and differences:

1. Higher prices than european calls

Since American options allow early exercise, their value is usually at least as high as European options. The difference is small in this case because calls are less likely to be exercised early, unlike puts.

2. Impact of correlation on prices

The stronger negative correlation (-0.70) results in a lower option price than the weaker negative correlation (-0.30). This is consistent with what we saw in questions 5 and 8—higher negative correlation increases volatility when prices drop, making call options cheaper.

3. Comparison to european call options

In question 5, the european call for correlation -0.30 was 2.89$, which is slightly higher than the american call at 2.70$.

In question 8, the european call for correlation -0.70 was 2.19$, again slightly higher than the american call at 2.02$.

This suggests that in this case, early exercise does not significantly increase the call price, reinforcing that American calls typically hold little early exercise value unless dividends are involved.

The results align with expectations—American calls are typically worth as much or slightly more than European calls. The correlation's impact remains similar to what we saw before: a stronger negative correlation reduces the call price due to increased volatility when prices fall.

Answer 14

Using the Heston model, we priced an Up-and-In call option with a $95 barrier, our result highlights how barrier conditions and correlation influence exotic option pricing.

the result gives

Correlation -0.30 → Price: 1.5638

Correlation -0.70 → Price: 1.5880

Observation:

Barrier Activation Effect: The option has no value unless the stock crosses the barrier before maturity. Its lower price, compared to a standard European call, reflects the reduced probability of activation.

Impact of Correlation: A stronger negative correlation (-0.7) results in a slightly higher price, indicating that greater stochastic volatility increases the likelihood of reaching the barrier, making the option marginally more valuable.

Comparison to European Call Options: Unlike a European call, which does not require a barrier activation, the up-and-in call is worth less due to the risk of never reaching the barrier. This price difference quantifies the probability of non-activation.

Answer 15

We also used the Merton model to price a down-and-in put option with a $65 barrier, our result gives:

Jump Intensity ($\lambda = 0.75$) → Price: 3.0115

Jump Intensity ($\lambda = 0.25$) → Price: 2.4836

Observations and Discussion:

Barrier Activation Effect. This option only holds value if the stock's price droped below $65 before maturity. And Its lower price compared to a standard European put reflects the chance that the barrier is never reached.

Impact of Jump Intensity: we realized that a higher jump intensity ($\lambda = 0.75$) increases the put price, as more frequent negative jumps raise the likelihood of activation.

When jump intensity is lower ($\lambda = 0.25$), the put is worth less because fewer sharp price drops reduce the probability of hitting the barrier.

Comparison to European Put Options: Unlike a European put, which has no activation requirement, the down-and-in put is cheaper due to the risk of non-activation. The price difference quantifies the probability that the stock never reaches the barrier.